Contents

Foreword by Stephen Hawking

Introduction by Frederick Seitz

Preface

what this book is about, and how to read it

Prologue: A Voyage among the Holes

in which the reader, in a science fiction tale, encounters black holes and all their strange properties as best we understand them in the 1990s

1. The Relativity of Space and Time

in which Einstein destroys Newton's conceptions of space and time as Absolute

2. The Warping of Space and Time

in which Hermann Minkowski unifies space and time, and Einstein warps them

3. Black Holes Discovered and Rejected

in which Einstein's laws of warped spacetime predict black holes, and Einstein rejects the prediction

4. The Mystery of the White Dwarfs